CHAPTER 6

Derivation, Variations and Optimization

Chapter overview

Objectives. Calculate a derivative on its domain; study variations and extrema; compare curves; optimize; exploit symmetries.
Prerequisites. Derivative number, algebraic calculation, trinomial sign, intervals.
Annual benchmark. December 1·10: 8:00 a.m., consolidation on 15 and 17

Diagnostic check: six questions, 10 minutes

Answer without consulting the course. Indicate the questions for which assistance is required. diagnostic check is not graded.

1. Derivative from x^2 in three?

2. Area of focus $\dfrac{1}{x-2}$?

3. Sign of ($x-1)(x+2)$?

4. Develop ($x^2+1)(x-2)$.

5. For $f(x)=x^2$, compare $f(-x)$ and $f(x)$.

6. If $f(1)=2$, $f\prime(1) = 0$, tangent equation?

After the diagnostic check. Retrieve the concepts indicated with the teacher; repeat a question and then explain the rule used.

Speaking and reasoning. Present a table of variations by distinguishing between the sign of f and its derivative. Explain why the bounds of an optimization problem need to be examined.

Search before formalizing

Can a increasing function have a horizontal tangent? Look for an example; can it Have a strictly negative derivative at an inner point?

Organization. Individual research 5 min, confrontation 2 5 min, pooling 5 min. Keep a trace of a test, even if not completed. A conjecture must then be justified.

6.1 Derivative function and usual functions

Definition and results

A function is detrievable over an interval when it is in each of its points; its derivative associates to x the number $f\prime(x)$ The usual derivatives are: constant $\mapsto 0$; $x \mapsto 1$; $x^2 \mapsto 2x$; $x^3 \mapsto 3x^2$; $\frac{1}{x} \mapsto -\frac{1}{x^2}$ for $x \neq 0$; $\sqrt{x} \mapsto \frac{1}{2\sqrt{x}}$ for $x > 0$. For $n \in \mathbb{Z}$, $(x^n)\prime = nx^{n-1}$ on its area of differentiability; n = 0, derivative is 0 directly.

Status and justification. Square: rate is 2 $a + h$. Inverse: for $a \neq 0$, $h \neq 0$ and $a + h \neq 0$, $\frac{[\frac{1}{a+h} - \frac{1}{a}]}{h} = -\frac{1}{[a(a+h)]}$, which tends to $-\frac{1}{a^2}$. The other formulas are allowed here.

Example solved. For $f(x) = x^{-3}$, the domain is $\mathbb{R} \setminus \{0\}$ and $f\prime(x) = -3x^{-4} = -\frac{3}{x^4}$. For $g(x) = \sqrt{x}$, the function exists in 0, its usual derivative only for $x > 0$.

Point of vigilance

The area of differentiability may be smaller than the area of definition.

Immediate application. Dirty x^5, x^{-2} and $\sqrt{x}$ specifying the areas.

6.2 Sum, product, inverse and quotient

Definition and results

For u and v derivables, ($u + v)\prime = u\prime + v\prime$, $(\lambda u)\prime = \lambda u\prime$, $(uv)\prime = u\prime v + uv\prime$. Where $v \neq 0$, $(\frac{1}{v})\prime = -\frac{v\prime}{v^2}$ and $(\frac{u}{v})\prime = \frac{u\prime v - uv\prime}{v^2}$.

Status and justification. For the product, add and cut $u(a + h)v(a)$ to the numerator. rate becomes $\frac{u(a+h)[v(a+h) - v(a)]}{h} + \frac{v(a)[u(a+h) - u(a)]}{h}$. As a drawable function is continuous (admitted), we get $u(a)v\prime(a) + v(a)u\prime(a)$ Other rules are allowed.

Example solved. For $f(x) = \frac{x^2+1}{x-1}$, $x \neq 1$, $f\prime(x) = \frac{[2x(x-1) - (x^2+1)]}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}$. Keeping parentheses avoids changing a single sign instead of the entire numerator.

Point of vigilance

The derivative of a product is not the product of the derivatives; the derivative of a quotient is not their quotient.

Immediate application. Derivative $(x^2+1)(3x-2)$ then $\dfrac{1}{x^2+1}$.

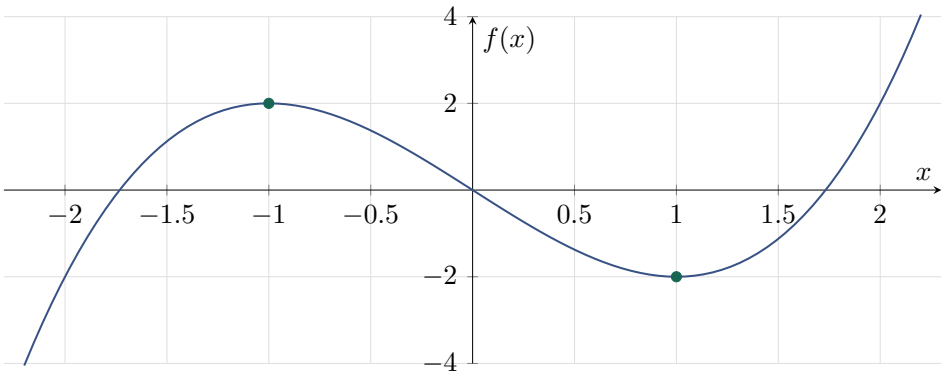
6.3 Sign of derivative and extrema

Definition and results

In an interval I, if $f' \geq 0$, f is increasing; if $f' \leq 0$, is decreasing; if $f' = 0$ everywhere, it is constant. A strictly positive derivative implies strict increase. If f has a local extremum at an inner point where it is detrievable, its derivative is zero. Reciprocality is false.

Status and justification. Theorems of variation allowed. For an interior minimum of a, rates on the right are positive or null, those on the left are negative or null; if they converge towards the same real, it Same argument for a maximum.

Example solved. $f(x) = x^3 - 3x$: $f'(x) = 3(x-1)(x+1)$. f grows on $]-\infty; -1]$, decreases on $[-1; 1]$, grows on $[1; +\infty[$; local maximum $f(-1) = 2$, local minimum $f(1) = -2$. None of these extrema n is global $\mathbb{R}$.



$f(x) = x^3 - 3x$: a local maximum in -1 and a local minimum in 1.

Point of vigilance

The theorem applies over an interval. A function may decrease over two intervals separated without decreasing on their meeting.

Immediate application. Study $f(x) = x^2 - 4x + 1$ on [0; 3] and give his extrema on this Interval.

6.4 Parity, Inequalities and Comparison

Definition and results

On a symmetric domain relative to 0, f is pairs if $f(-x) = f(x)$ and odd if $f(-x) = -f(x$The curve of a pair function is symmetrical in relation to the axis of the
In order to compare f and g, the sign is studied
of $f - g$, possibly with its derivative.

Status and justification. The symmetries are derived from the coordinates of the image points. Inequality, a study of variations can give a minimum of the difference, then its sign; the derivative does not substitute directly for this difference.

Example solved. For $x > 0$, $h(x) = x + \frac{1}{x} - 2$a for derivative $\frac{x^2-1}{x^2}$. It decreases up to 1 then grows; $h(1) = 0$, therefore $x + \frac{1}{x} \geq 2$. Equality is achieved only in 1.

Point of vigilance

Verify the domain before parity. $f\prime - g\prime$ information on the change in $f - g$, not directly on his sign.

Immediate application. Studying gender parity $x^4 + x^2$ and $x^3 - x$; Demonstrate $x^2 + 1 \geq 2x$.

6.5 Newton Optimization and Method

Definition and results

Optimize is to define the variable and its domain, express the size to optimize, then compare all the candidate values, including the bounds.

uses tangent to approach a solution of $f(x) = 0 : x_{n+1} = x_n - \frac{f(x_n)}{f}\prime(x_n)$ where $f\prime(x_n) \neq 0$. It is used only in favourable cases.

Status and justification. The Newton formula comes from the intersection with the horizontal axis of $y = f(x_n) + f\prime(x_n)(x - x_n)$ Convergence is not guaranteed in general; it is recognized and monitored in the case chosen.

Example solved. For $f(x) = x^2 - 2$ with $x_0 = 1$, Newton gives $x_1 = \frac{3}{2}$, $x_2 = \frac{17}{12}$. Squares are 2.25 then about 2,00694 : we are approaching $\sqrt{2}$. The program must control derivative zero and number maximum steps.

Point of vigilance

A numerical value provided by an algorithm is not proof of minimum or convergence.

Immediate application. A perimeter rectangle 12 m wide x. Determine maximum area by diversion.

Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 6.1 · A collection of derivatives

Functions are considered

$$f(x) = 3x^4 - 2x + 7, \quad g(x) = (x^2 + 2)(x - 1), \quad h(x) = \frac{2x + 1}{x + 3}.$$

1. Specify the area of definition of each of these functions.
2. Calculate their derivatives by naming each time the rule used. g, first keep the two terms of the product rule; for h, write the numerator before reducing it.
3. Control the result concerning g developing its expression before drifting.

Exercise 6.2 · A complete picture

The function f is defined on [0; 4] by $f(x) = x^3 - 6x^2 + 9x$ + 1. We look for its minimum values and maximum in this interval.

1. Calculate and factorize $f\prime(x)$, then study his sign on [0; 4].
2. Draw up the table of variations of the f indicating the values at the bounds and at the points where the Derivative from the void.
3. Give the minimum and maximum of f on the interval, as well as all the apses where they are Affected.

Exercise 6.3 · A quotient

For any real strictly positive x, we're laying $f(x) = x + \frac{4}{x}$.

1. Studying variations in f on]0; + ∞[. Present the factors that determine the sign of derivative.
2. Deduct the smallest value of $x + \frac{4}{x}$, then write an inequality valid for all $x > 0$. Specify exactly the case of equality.

Exercise 6.4 · Compare two curves

Functions $f(x) = x^3$ and $g(x) = x$ are defined on $\mathbb{R}$. We note $\mathcal{C}_f$ and $\mathcal{C}_g$ their curves in a same coordinate system.

1. Determine the intersection points of these two curves.

2. Study their relative position on $\mathbb{R}$ by choosing a suitable method. Translate your result by intervals where $\mathcal{C}_f$ is above or below $\mathcal{C}_g$.

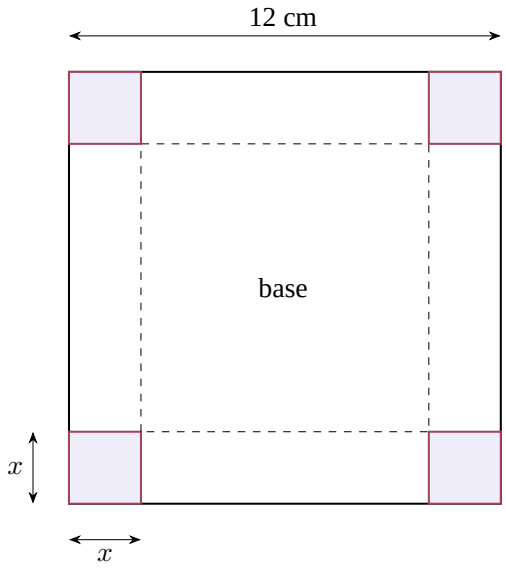
Exercise 6.5 · One package

A box without lid is made from a square sheet of 12 cm side.
in each of the four corners a square side x cm, then the remaining edges are raised.
The leaf is neglected.

1. Draw the pattern and express the height and the two dimensions of the base of the box. Specify the values of x which give a non-zero volume box.

2. Express Volume $V(x)$ in cubic centimetres. Determine the dimensions that make this volume maximum.

3. Check the values at the two bounds of the domain and give the exact maximum volume.



The four coloured areas are removed; the dots indicate the folds. The drawing represents a illustrative value of x and does not give the optimum.

Exercise 6.6 · Stationary without extremum

A student says: "If the derivative cancels in one point, the function necessarily has an extreme
"We are studying the function. $f(x) = x^3$ to test this statement.

1. Calculate $f\prime(0)$.

2. Compare to $f(0)$ values of $f(x)$ for real x negative then positive, as close to 0 as We want to.

3. Conclude on the existence of a local extremum in 0 and on the proposed statement.

Exercise 6.7 · Compare Strategies

We want to show that, for all real x, $x^2 - 6x + 12 \geq 3$.

1. Propose a demonstration by showing a square.

2. Propose a second demonstration by studying variations in a function.

3. Specify the case of equality. Compare the two methods: which one asks here the least calculation?

Fiscal year 6.8 · Deferred review

Reciprocal function $f(x) = \dfrac{1}{x}$ a for field $D = \mathbb{R} \setminus \{0\}$. A student calculates $f\prime(x) = -\dfrac{1}{x^2} < 0$ and concludes that f is decreasing on all D.

1. Identify the missing hypothesis in the application of the theorem linking derivative sign and changes.

2. Indicate the two intervals on which its conclusion is correct.

3. Using Numbers -1 and 1, build a counter-example to its conclusion on all D.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, Exercise 12 p. 156, Questions 1 and 2. Calculating the difference, studying its sign and translate it into the relative position of the curves.

CQFD, exercise 16 p. 156, questions f and g. Give an interval for each function where it is both negative and increasing; justify both properties separately.

MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $(x^3)\prime$ value
   A. x^2
   B. $3x^2$
   C. $3x$
   D. $\frac{x^4}{4}$
2. $(\frac{1}{x})\prime$ value
   A. $-\frac{1}{x^2}$
   B. $\frac{1}{x^2}$
   C. $-\frac{1}{x}$
   D. 1
3. $(uv)\prime$ value
   A. $u\prime v\prime$
   B. $u\prime + v\prime$
   C. $uv\prime$
   D. $u\prime v + uv\prime$
4. $f\prime > 0$ over an interval implies
   A.f. positive
   B. f. negative
   C. f. Strictly increasing
   D. f. constant
5. $\frac{x^2+1}{x-1}$ is defined if
   A. $x > 0$
   B. $x \neq 1$
   C. $x \neq -1$
   D. $x \neq 0$

MCQ 2: understand and choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $x^3 - x$ is

   A. Pair

   B. constant

   C. odd

   D. Neither pair nor odd

2. $f\prime(a)$ = 0 necessarily implies

   A. Horizontal tangent

   B. Maximum

   C. Minimum

   D. f. constant

3. Derivative from $\sqrt{x}$:

   A. $2\sqrt{x}$

   B. $\dfrac{1}{\sqrt{x}}$

   C. $\dfrac{\sqrt{x}}{2}$

   D. $\dfrac{1}{2\sqrt{x}}, x > 0$

4. Minimum of $x + \dfrac{1}{x}$ on]0; + ∞[:

   A. 0

   B. 2

   C. 1

   D. None

5. Newton for $x^2 - 2$ is not defined with $x_0 =$

   A. 1

   B. 2

   C. 0

   D. -1

True or false?

1. The derivative of a positive function is positive.

2. An inner extremum of a drifting function gives a zero derivative.

3. The output of two increasing functions is increasing.

Personal review

I drift on the right domain; I simplify without losing the forbidden values; I distinguish extremum
local and global; I sometimes choose factorization rather than derivative.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and
a mistake to avoid.

Corrected and conduct of chapter 6

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. 6. *Action:* Resuming rate 2 $a+h$.

2. $\mathbb{R}\setminus\{2\}$. *Action:* Cancel denominator.

3. Positive outside of [-2; 1], negative between roots, zero at bounds. *Action:* Build two lines of signs.

4. x^3-2x^2+x-2. *Action:* Distribute each term without omission.

5. They are equal. *Action:* Replace -x in parentheses.

6. $y=2$. *Action:* Resuming the tangent equation and simplifying.

Research activity

x^3 is strictly increasing with horizontal tangent in 0. For an increasing function derivatable, the rates on both sides are positive or zero, so the derivative cannot be negative.

Immediate applications

1. $5x^4$ on $\mathbb{R}$; $-\dfrac{2}{x^3}$ for $x\neq 0$; $\dfrac{1}{2\sqrt{x}}$ for $x>0$.

2. $2x(3x-2)+3(x^2+1)=9x^2-4x+3$; $-\dfrac{2x}{(x^2+1)^2}$ on $\mathbb{R}$.

3. $f\prime=2x-4$, so decrease to 2 then increase. Minimum -3 in 2 ; values at bounds 1 and -2, so maximum 1 in 0.

4. First pair, second odd; difference $x^2+1-2x=(x-1)^2\geq 0$, equal to 1.

5. $A(x)=x(6-x)$ on [0; 6] ; $A\prime=6-2x$. Maximum 9 m ^{2} in 3.

Exercises: approach and results

Exercise 6.1 · A collection of derivatives. $12x^3-2$; $2x(x-1)+(x^2+2)=3x^2-2x+2$; $\dfrac{[2(x+3)-(2x+1)]}{(x+3)^2}=\dfrac{5}{(x+3)^2}$ for $x\neq -3$.

Exercise 6.2 · A complete picture. $f\prime=3(x-1)(x-3)$; signs + , −, +. Values $f(0)=1$, $f(1)=5$, $f(3)=1$, $f(4)=5$. Minimum 1 in 0 and 3; maximum 5 in 1 and 4.

Exercise 6.3 · A quotient. $f\prime=1-\dfrac{4}{x^2}=\dfrac{(x-2)(x+2)}{x^2}$. Negative sign before 2, positive after. Minimum 4 in 2 ; for $x>0$, $x+\dfrac{4}{x}\geq 4$, equality only in 2.

Exercise 6.4 · Compare two curves. $f-g=x(x-1)(x+1$. Signs: negative on] $-\infty;-1[$, positive about] -1; 0[, negative on]0; 1[, positive on]1; + ∞[. Intersection in $-1,0,1$. One can do the table of three factors without derivation.

Exercise 6.5 · One package. $V=x(12-2x)^2=4x(6-x)^2$. $V'=12(x-2)(x-6)$, positive on [0; 2[, negative on]2; 6[. $V(0)=V(6)=0$, V(2) = 128 cm ^{3}. Maximum for x=2 cm; box 8 $\times 8\times 2$.

Exercise 6.6 · Stationary without extremum. $f'=3x^2$, so value 0 in 0. In any vicinity of 0, negative x give $f(x)<0$ and positive x $f(x)>0$; no extremum.

Exercise 6.7 · Compare strategies. Square: $(x-3)^2+3\geq 3$. Derivative: $2x-6$, minimum in 3, value 3. The first avoids a complete study of variation; the second illustrates the link between Chapters.

Fiscal year 6.8 · Deferred review. It decreases on] $-\infty$; 0[and]0; +∞[. It does not decrease on their meetings: $-1<1$ but $f(-1)=-1<1=f$(1). The domain is not an interval.

CQFD, Exercise 12 p. 156, Questions 1 and 2. $f(x)-g(x)=\frac{1}{2}x^2-1=\frac{1}{2}(x-\sqrt{2})(x+\sqrt{2})$. The f curve is strictly above that of g for $x<-\sqrt{2}$ or $x>\sqrt{2}$, below between these two numbers; equality to the apses $\pm\ \overline{2}$.

CQFD, exercise 16 p. 156, questions f and g. For $f(x)=x^3$, we can choose] $-\infty;0[$: $f(x)<0$ and the function is strictly increasing. $g(x)=x^2-4$, one can choose [0; 2[: $g(x)<0$ and the function is strictly increasing over this interval. Sub-intervals not reduced to one point Also suitable.

MCQ: answers and reasons

MCQ 1.

1. B. Whole power.

2. A. Inverse rate.

3. D. Product rule.

4. C. Sign of derivative and variation.

5. B. Non-zero denominator.

MCQ 2.

1. C. $f(-x)=-f(x)$.

2. A. The other conclusions call for more.

3. D. Formula and field.

4. B. Achieved in 1.

5. C. Derivative $2x_0=0$.

True or false

1. False: $f(x) = \dfrac{1}{x} > 0$ on]0; + ∞[, but $f\prime(x) < 0$.

2. True, with the assumptions specified; at the bounds, the conclusion is not imposed.

3. False without sign assumptions: on [$-2; -1]$, $u(x) = v(x) = x$ growth, but x^2 Decline.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.